Math Class 11 FBISE

Chapter # 3 Vectors

MOST IMP. Questions

Wed

i) FLP

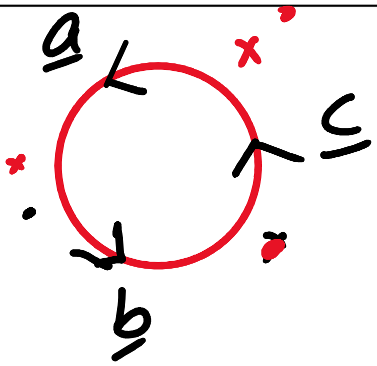
+ Formula sheet

Thursday

'MCQs'

$3\times4 + 1\times8 + 3$ MCQs

If $\underline{a} = 2\underline{i} - \underline{j} + 3\underline{k}$, $\underline{b} = 3\underline{i} + 2\underline{j} + 4\underline{k}$ and $\underline{c} = \underline{i} + 3\underline{j} - 5\underline{k}$, then verify that $\underline{a} \cdot \underline{b} \times \underline{c} = \underline{b} \cdot \underline{c} \times \underline{a} = \underline{c} \cdot \underline{a} \times \underline{b}$.	8



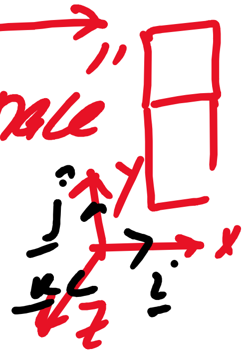
$\underline{a} \cdot \underline{b} = |a||b| \cos\theta$

'scalar'

$|\underline{a} \cdot \underline{b} \times \underline{c}| =$ Volume of $\boxed{\text{Parallelopipe}}$

$\frac{1}{6}|\underline{a} \cdot \underline{b} \times \underline{c}| =$ Volume of Tetrahedron

"space"



If $\underline{a} = 2\underline{i} - \underline{j} + 3\underline{k}$, $\underline{b} = 3\underline{i} + 2\underline{j} + 4\underline{k}$ and $\underline{c} = \underline{i} + 3\underline{j} - 5\underline{k}$, then verify that $\underline{a} \cdot \underline{b} \times \underline{c} = \underline{b} \cdot \underline{c} \times \underline{a} = \underline{c} \cdot \underline{a} \times \underline{b}$.	8

$$\underline{a}\cdot\underline{b}\times\underline{c} = \begin{vmatrix} 2 & -1 & 3 \\ 3 & 2 & 4 \\ 1 & 3 & -5 \end{vmatrix}$$

$(-1)^{1+2}$

$$= 2(-10-12)+1(-15-4)+3(9-2)$$

$$= -44-19+21 = -42 \text{ cubic units}$$

$$|\underline{a}\cdot\underline{b}\times\underline{c}| = |-42| = 42 \text{ cubic units}$$

$$|\underline{b}\cdot\underline{c}\times\underline{a}| = 42$$

$$|\underline{c}\cdot\underline{a}\times\underline{b}| = \underline{\underline{42}}$$

Volume of parallelopipe,

copy + pen

Volume of Tetrahedron $= \frac{1}{6}\left|\underline{a}\cdot\underline{b}\times\underline{c}\right|$

Top the list

A force $\vec{F} = 3\underline{i} - 2\underline{j} + 5\underline{k}$ acts on a particle at point $P\ (3, -4, 2)$. Find moment of the force about origin and a point $(1, -1, -1)$.	4

i) Torque
↳ Moment of Force / Torque

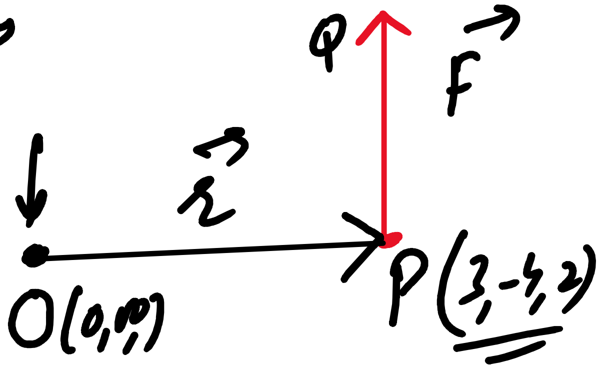
ii) Work Done

iii) Volume of Tetra... Vol

iv) Area of parallelogram

v) Area of triangle

$\vec{\tau} = \vec{r} \times \vec{F}$



$\vec{\tau} = \vec{OP} \times \vec{PQ}$

$\vec{\tau} = \vec{r} \times \vec{F} \rightarrow ①$

$$\vec{OP} = 3i - 4j + 2k$$

$$\vec{r} = \begin{vmatrix} i & j & j \\ 3 & -4 & 2 \\ 3 & -2 & 5 \end{vmatrix}$$

$A \times B =$

$$\vec{r} = i(-20+4) - j(15-6) + k(-6+12)$$

$\vec{r} = 3i - 4j + 2k$

$$\vec{r} = -16i - 9j + 6k$$

$\vec{F} = 3i - 2j + 5k$

Check

$\vec{r} \cdot \vec{r} = -48 + 36 + 12 = 0$

$\vec{r} \cdot \vec{F} = -48 + 18 + 30 = 0$

A force $\vec{F} = 3\underline{i} - 2\underline{j} + 5\underline{k}$ acts on a particle at point P $(3, -4, 2)$. Find moment of the force about origin and a point $(1, -1, -1)$.	4

A

$R(3,-4,2)$ $\vec{OP}$ O

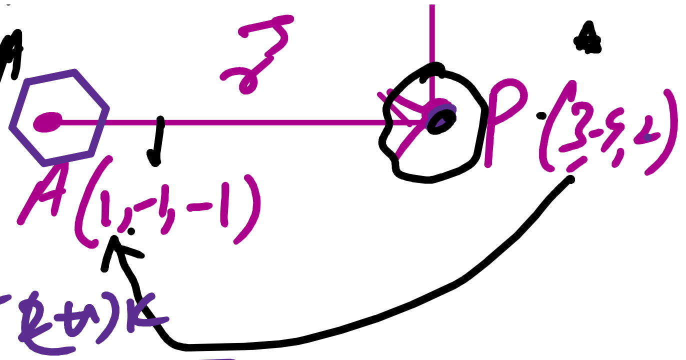
Q $\vec{F}$

$-4-(-1)$

b) $\vec{OP} - \vec{OA}$

b)

Moment arm $= \vec{r} = \vec{AP} = \vec{OP} - \vec{OA}$

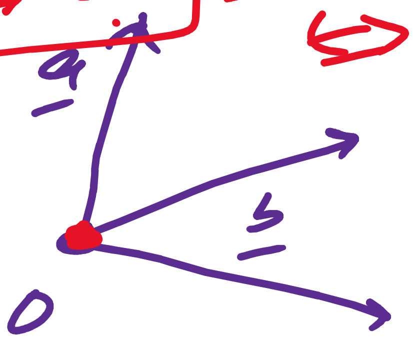


$\vec{r} = (3-1)\underline{i} + (-4+1)\underline{j} + (2+1)\underline{k}$

$$\boxed{\vec{r} = 2\underline{i} - 3\underline{j} + 3\underline{k}}$$

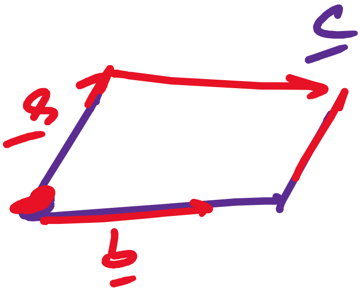
$\boxed{\vec{r} \times \vec{F} = -\vec{F} \times \vec{r}}$ $\quad \vec{\Gamma} = \vec{r} \times \vec{F} = \begin{vmatrix} i & j & k \\ 2 & -3 & 3 \end{vmatrix}$

Find volume of the tetrahedron if $a = 2\underline{i} - 3j + \underline{k}$, $b = \underline{i} + 2j - \underline{k}$ and $c = -3\underline{i} - \underline{j} + 5\underline{k}$ are its coterminous edges.	4



Vol of Tetrahedron $= \frac{1}{6} \left| \underline{a} \cdot \underline{b} \times \underline{c} \right|$

$$= \frac{1}{6} \begin{vmatrix} 2 & -3 & 1 \\ 1 & 2 & -1 \\ -3 & -1 & 5 \end{vmatrix}$$



If the angle between two vectors $\underline{a} = 2\underline{i} - 3\underline{j} + 4\underline{k}$ and $\underline{b} = \underline{i} + 2\underline{j} + 2\underline{k}$ is θ, then find the values of $\cos\theta$ and $\sin\theta$.	4

11 90° $|\underline{a}|\,|\underline{b}|$

$\underline{i}\cdot\underline{i} = 1$, $\underline{i}\cdot\underline{j} = 0$

sol

$$\underline{a}\cdot\underline{b} = a\,b\cos\theta$$

$$\cos\theta = \frac{\underline{a}\cdot\underline{b}}{|\underline{a}|\,|\underline{b}|} \rightarrow (1)$$

$$\underline{a}\cdot\underline{b} = 2 - 6 + 8 = 4$$

$$|\underline{a}| = \sqrt{(2)^2 + (-3)^2 + (4)^2}$$

$$|\underline{a}| = \sqrt{4 + 9 + 16} = \sqrt{29}$$

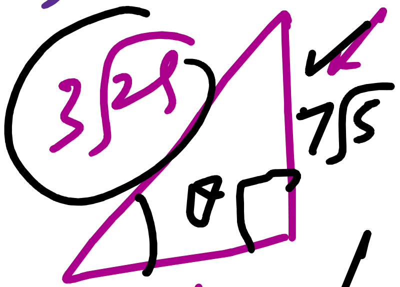
$$|\underline{b}| = \sqrt{1 + 4 + 4} = \sqrt{9} = 3$$

$$(1)\quad \cos\theta = \frac{4}{\sqrt{29} \times 3} = \frac{4}{3\sqrt{29}}$$

$$\theta = \cos^{-1}\left(\frac{4}{3\sqrt{29}}\right)$$

$$\sin^2\theta = 1 - \cos^2\theta$$

$$\sin\theta = \sqrt{1 - \cos^2\theta}$$



$\sin\theta = 7\sqrt{5}$

$\sin\alpha = \sqrt{1-\cos^2\alpha}$

$= \sqrt{1 - \left(\frac{4}{3\sqrt{28}}\right)^2}$

$\boxed{\sin\alpha = \frac{7\sqrt{5}}{3\sqrt{28}}}$

$\sin\alpha = \frac{7\sqrt{45}}{87}$

Find the value of m if vectors $\underline{a} = \hat{\imath} + m\hat{\jmath} + 2\hat{k}$, $\underline{b} = 2\hat{\imath} - \hat{\jmath} + 3\hat{k}$, $\underline{c} = 3\hat{\imath} + 2\hat{\jmath} + \hat{k}$ are coplanar.	04

Coplaner

$\boxed{\underline{a} \cdot \underline{b} \times c = 0}$

$\underline{a} \cdot \underline{b} \times \underline{c} = \begin{vmatrix} 1 & m & 2 \\ 2 & -1 & 3 \\ 2 & 2 & 1 \end{vmatrix} = 0$

$$a \cdot b \times c = \begin{vmatrix} \\ 3 & 2 & 1 \end{vmatrix} = 0$$

$$= 1(-1-6) - m(2-9) + 2(4+3)$$

$$-7 + 7m + 14 = 0$$

$$7m = -7$$

$$\boxed{m = -1}$$

$$1)\ a\cos\theta = \frac{5-4+2}{\sqrt{1+4+4}} = \frac{3}{3} = 1$$

$$i)\ b\cos\alpha = \frac{a \cdot b}{|a|} = \frac{3}{\sqrt{25+4+1}} = \frac{3}{\sqrt{30}}$$

If $\underline{a} = 5\hat{\imath} - 2\hat{\jmath} + \hat{k}$ and $\underline{b} = \hat{\imath} + 2\hat{\jmath} + 2\hat{k}$, then find projection of $\underline{a}$ along $\underline{b}$ and projection of $\underline{b}$ along $\underline{a}$.	04

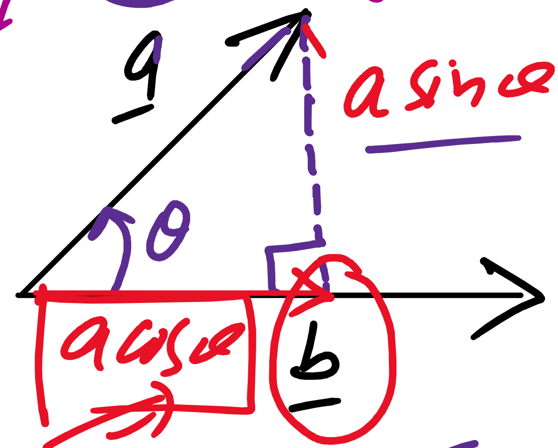
dot product b/w two vectors

dot product ...

$$\underline{a}\cdot\underline{b} = |\underline{a}||\underline{b}|\cos\theta$$

a) Projection of $\underline{a}$ along $\underline{b}$

$a\cos\theta$??



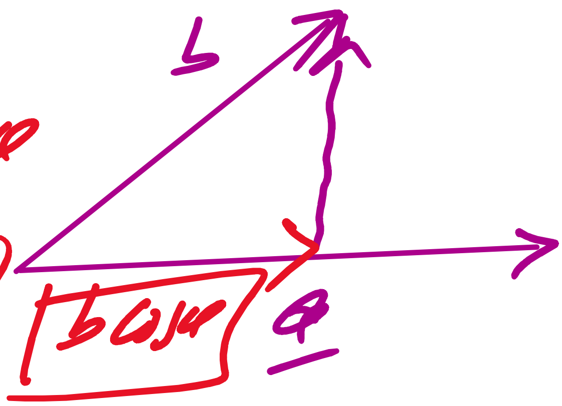
$$|a|\cos\theta = \frac{\boxed{\vec{a}\cdot\vec{b}}}{|\underline{b}|}$$

$\underline{v}$, $\underline{u}$

Projection of $\underline{b}$ along $\underline{a}$

$$\underline{a}\cdot\underline{b} = a\,b\cos\theta$$

$$\Rightarrow \boxed{b\cos\theta = \frac{a\cdot b}{a}}$$



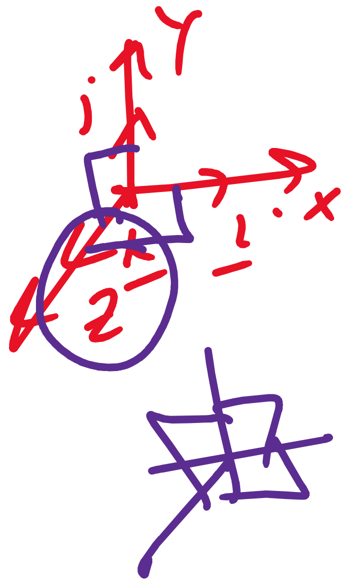
$\underline{a} = \hat{i} + 0\hat{j} + 0\hat{k}$

Apply cross product to find angle between two vectors $\underline{a} = \hat{\imath}$ and $\underline{b} = \frac{1}{2}\hat{\imath} + \frac{\sqrt{3}}{2}\hat{\jmath} + 0k$	04

$$\underline{a} \times \underline{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \end{vmatrix}$$

$$\underline{a} \times \underline{b} = \hat{i}(0) - \hat{j}(0) + \underline{k}\left(\frac{\sqrt{3}}{2}\right)$$

$$\boxed{\underline{a} \times \underline{b} = \frac{\sqrt{3}}{2}\,\underline{k}}$$



$|\hat{n}| = 1$

$$\left|a\, b \sin\theta\, \hat{n}\right| = \left|\frac{3}{2}\,\hat{k}\right|$$

$$a\, b \sin\theta = \frac{3}{2} \rightarrow (1)$$

$\vec{a} \times \vec{b}$

$$a = 1, \quad b = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$$

$$\boxed{|a = 1}, \quad b = \sqrt{\frac{1}{4} + \frac{3}{4}}$$

$$\boxed{|b = 1}$$

$|b| = 1$

$① \Rightarrow \sin\theta = \frac{\sqrt{3}}{2}$

$\theta = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = 60°$

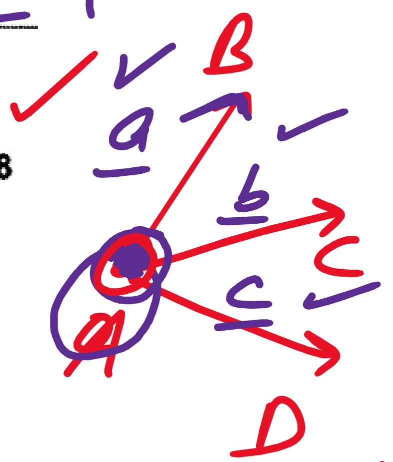
$1-3$ $\quad$ $\vec{AB} = \vec{OB} - \vec{OA}$ $\quad$ $|\equiv|$

The vertices of a tetrahedron are $A(1,2,3), B(2,31), C(3,1,2)$ and $D(4,2,1)$

a. Find volume of the tetrahedron $ABCD$

b. Using same vectors, find the volume of the parallelepiped formed by $\overrightarrow{AB}$, $\overrightarrow{AC}$ and $\overrightarrow{AD}$.

08



$\vec{AD} \cdot (\vec{AC} \times \vec{AD})$

a) $\vec{a} = \vec{AB} = \hat{i} + \hat{j} - 2\hat{k}$

$\vec{b} = \vec{AC} = 2\hat{i} - \hat{j} - \hat{k}$

$\vec{c} = \vec{AD} = 3\hat{i} + 0\hat{j} - 2\hat{k}$

Volume of Tet $= \frac{1}{6}|\vec{a} \cdot \vec{b} \times \vec{c}|$

$= \dfrac{1}{6} \begin{vmatrix} 1 & 1 & -2 \\ +2 & -1 & -1 \\ 3 & 0 & -2 \end{vmatrix}$

$|\vec{a} \cdot \vec{b} \times \vec{c}|$

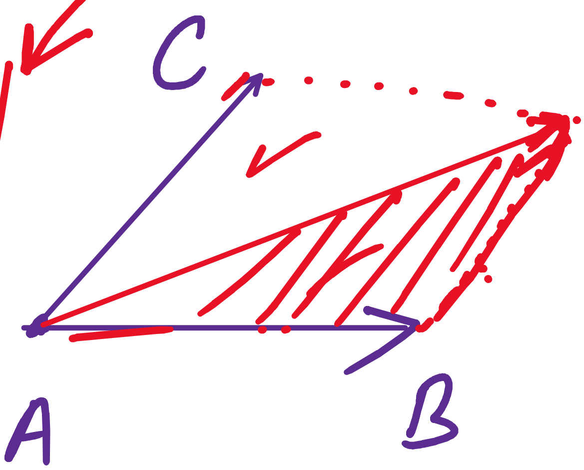
c) $|a \cdot b \times c|$

$= \Big|$ $\Big|$

A triangular park has its three corners located at points **A** (2, 3, 1), **B** (5, 7, 2), and **C** (1, 4, 3). Find area of the park using cross product of vectors.	**04**

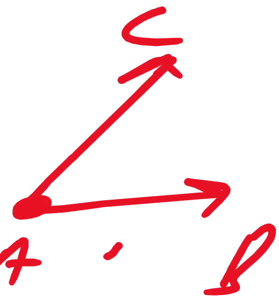
$|\vec{AB} \times \vec{AC}|$

$=$



Area of triangle $= \frac{1}{2}|\vec{AB} \times \vec{AC}|$

A triangular park has its three corners located at points **A** (2, 3, 1), **B** (5, 7, 2), and **C** (1, 4, 3). Find area of the park using cross product of vectors.	**04**



Area of Triangular Park $= \frac{1}{2}|\vec{AB} \times \vec{AC}| \rightarrow (1)$

$\vec{AB} = 3\hat{i} + 4\hat{j} + \hat{k}$

$\vec{AC} = -\hat{i} + \hat{j} + 2\hat{k}$

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 4 & 1 \\ -1 & 1 & 2 \end{vmatrix}$$

$=$

Find the value of p such that vectors $\hat{i}+2\hat{j}+p\hat{k}$, $3\hat{i}+p\hat{j}+4\hat{k}$ and $2\hat{i}+3\hat{j}+4\hat{k}$ are coplanar.

$$|\underline{a}\cdot\underline{b}\times\underline{c}| = 0$$

$$= \begin{vmatrix} 1 & 2 & p \\ 3 & p & 4 \\ 2 & 3 & 4 \end{vmatrix} = 0$$

$$1(4p-12)-2(12-8)+p(9-2p) = 0$$

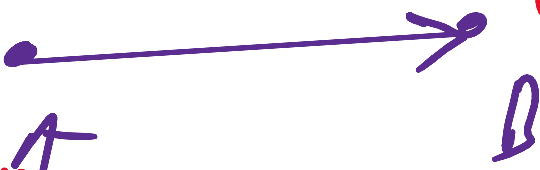
A force $\vec{F} = 6\hat{i} + 8\hat{j} + 4\hat{k}$ acts on an object. The object moves from A (1,2,3) to point B(4,6,5).
(a) Find the displacement vector $\vec{d}$.
(b) Calculate work done by the force $\vec{F}$.

~~d~~ $\boxed{W = \vec{F} \cdot \vec{d}}$

$(\quad)\cdot(\quad)$

a)

$d = \vec{AB}$



$\vec{d} = 3\hat{i} + 4\hat{j} + 2\hat{k}$

b) $W = \vec{F} \cdot \vec{d}$

$= 18 + 32 + 8 = 58$ un

If $\vec{a} = 3\hat{i} - 2\hat{j} + 4\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$, then

(a) Find $\vec{a} \times \vec{b}$.

(b) Find $\left(\vec{a} \times \vec{b}\right) \cdot \vec{a}$ and verify that $\vec{a} \times \vec{b} \perp \vec{a}$.

(c) Find $\left(\vec{a} \times \vec{b}\right) \cdot \vec{b}$ and verify that $\vec{a} \times \vec{b} \perp \vec{b}$

08

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -2 & 4 \\ 1 & 2 & -1 \end{vmatrix}$$

$$a \times b = -6\hat{i} + 7\hat{j} + 8\hat{k}$$

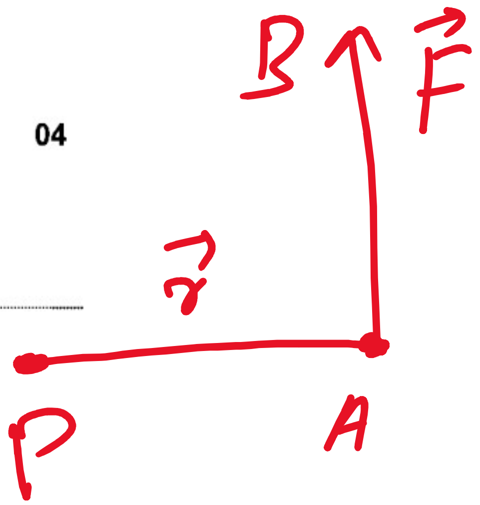
$$3\hat{i} - 2\hat{j} + 4\hat{k}$$

$$(\vec{a} \times \vec{b}) \cdot \vec{a} = -18 - 14 + 32$$

$$0$$

$-3-(-2) = -3+2 = -1$

A mechanic applies a force $\vec{F} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ at point $A\,(14, -3,\ 6)$ to turn a bolt with pivot at $P\,(2, -2, 1)$. Find: **(a)** the moment arm $\vec{r}$ and **(b)** the moment vector $\overrightarrow{M}$ about the bolt.	**04**



$\vec{r} = \vec{PA} = 12\hat{i} - \hat{j} + 5\hat{k}$

Find the value of λ such that vectors $\hat{i}+4\hat{j}+2\hat{k}$, $2\hat{i}-\hat{j}+3\hat{k}$ and $\lambda\hat{i}+3\hat{j}+5\hat{k}$ are coplanar.

$$\underline{a} \cdot \underline{b} \times \underline{c} = 0$$

$$\begin{vmatrix} 1 & 4 & 2 \\ 2 & -1 & 3 \\ \lambda & 3 & 5 \end{vmatrix} = 0$$

A force $\vec{F}=5\hat{i}+3\hat{j}+4\hat{k}$ is applied to move an object from $A(2,3,1)$ to $B(5,7,4)$. Find work done by the force.	**04**

$\vec{d}=\vec{AB}=3\hat{i}+4\hat{j}+3\hat{k}$

$W=\vec{F}\cdot\vec{d}=$

If $\vec{a} = 5\hat{i} - 3\hat{j} + \hat{k}$ and $\vec{b} = -2\hat{i} + 6\hat{j} + 4\hat{k}$, then

(a) Find $\vec{a} \times \vec{b}$

(b) Find $\left(\vec{a} \times \vec{b}\right) \cdot \vec{a}$ and verify that $\vec{a} \times \vec{b} \perp \vec{a}$ 08

(c) Find $\left(\vec{a} \times \vec{b}\right) \cdot \vec{b}$ and verify that $\vec{a} \times \vec{b} \perp \vec{b}$

Example 4: } ⇒ 2nd //

Find the vector which divides $\overrightarrow{OA} = \underline{a} = 3\hat{\imath} + \hat{\jmath} + 10\hat{k}$ and $\overrightarrow{OB} = \underline{b} = 3\hat{\imath} + 7\hat{\jmath} + \hat{k}$ in the ratio 3: 2(Internally)

$\vec{r} = ?$

Sol Given $m : n = 3 : 2$

$2 : 3$

$m = 3$ $\quad \underline{a} = 3\hat{i} + \hat{j} + 10\hat{k}$

$n = 2$ $\quad \underline{b} = 3\hat{i} + 7\hat{j} + \hat{k}$

$$\vec{r} = \overrightarrow{OP} = \frac{m\underline{b} + n\underline{a}}{m+n} \rightarrow (1)$$

$$= \frac{3(3\hat{i} + 7\hat{j} + \hat{k}) + 2(3\hat{i} + \hat{j} + 10\hat{k})}{3+2}$$

Find the vector which divides $\overrightarrow{OA} = \underline{a} = 3\hat{\imath} + \hat{\jmath} + 10\hat{k}$ and $\overrightarrow{OB} = \boxed{\underline{b} = 3\hat{\imath} + 7\hat{\jmath} + \hat{k}}$ in the ratio 3:2(externally)

mb

$m \quad n$

$$\boxed{\vec{r} = \vec{OP} = \frac{m\underline{b} - n\underline{a}}{m - n}}$$

✓

For Any doubts, email me

fakhar.abbas@nu.edu.pk.

4. (i) Find the value of m for which the vector $\vec{a} = 3\hat{\imath} + 4\hat{\jmath} - 9\hat{k}$ is parallel to $\vec{b} = \hat{\imath} + m\hat{\jmath} - 3\hat{k}$.

(ii) Find the value of λ for which the points P, Q and R are collinear. Given that $\hat{\imath} + 2\hat{\jmath} + 3\hat{k}, -2\hat{\imath} + 2\hat{\jmath} + 6\hat{k}$ and $\lambda\hat{\imath} - \lambda\hat{k}$ are the position vectors of points P, Q and R respectively.

i) Since vectors parallel vectors

Sol)

$$\frac{3}{1} = \frac{4}{m} = \frac{-9}{-3}$$

$3 = \boxed{\frac{4}{m} = 3}$

$\perp \quad \parallel \quad \Rightarrow \quad \frac{4}{m} = 3 \Rightarrow m = \frac{4}{3}$

$\hat{a} \parallel \hat{b} \Rightarrow \cancel{\vec{a} = \lambda \vec{b}}$

$a = \lambda b$

$3\hat{i} + 4\hat{j} - 9\hat{k} = \lambda(\hat{i} + m\hat{j} + 3\hat{k})$

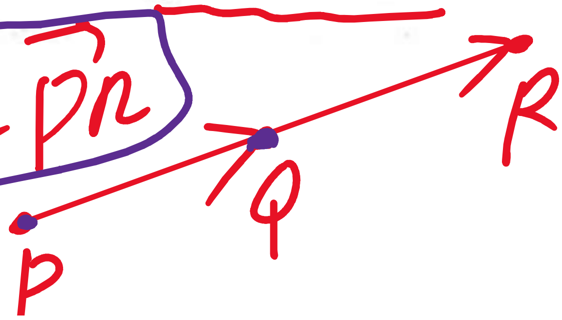
$3\hat{i} + 4\hat{j} - 9\hat{k} = \lambda\hat{i} + m\lambda\hat{j} + 3\lambda\hat{k}$

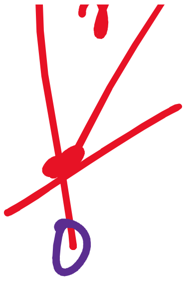
$\boxed{\lambda = 3} \Rightarrow m\lambda = 4$

$3m = 4 \Rightarrow m = 4/3$

Find the value of λ for which the points P, Q and R are collinear.
Given that $\hat{\imath} + 2\hat{\jmath} + 3\hat{k}, -2\hat{\imath} + 2\hat{\jmath} + 6\hat{k}$ and $\lambda\hat{\imath} - \lambda\hat{k}$ are the position vectors of points P, Q and R respectively.

$\vec{r}$ P $\boxed{\overrightarrow{PQ} + \overrightarrow{QR} = \overrightarrow{PR}}$





Sol

$\vec{OP} = \hat{i}+2\hat{j}+3\hat{k}$ ✓

$\vec{OQ} = -2\hat{i}+2\hat{j}+6\hat{k}$ ✓

$\vec{OR} = 1\hat{i}+0\hat{j}+1\hat{k}$

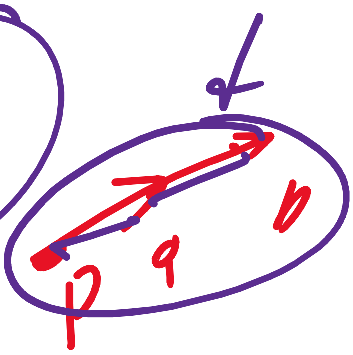
$\vec{PQ} = \vec{OQ} - \vec{OP}$

$\vec{PQ} = -3\hat{i}+0\hat{j}+3\hat{k}$

$\vec{QR} = \vec{OR} - \vec{OQ}$

$\vec{PR} =$

$\vec{PQ}, \vec{QR}, \vec{PR}$



$-3 = \frac{\;}{1}$

Short Q

5. (i) If $\vec{a} = \hat{\imath} - 2\hat{\jmath} + \hat{k}$; $\vec{b} = \hat{\imath} - \hat{\jmath} - \hat{k}$ and $\vec{c} = 2\hat{\imath} + \hat{k}$ then find a unit vector in the direction of $2\vec{a} - 3\vec{b} + \vec{c}$.

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(ii) Use vectors to find the length of diagonals of a parallelogram having adjacent sides $\hat{\imath} + \hat{\jmath}$ and $\hat{\imath} - 2\hat{\jmath}$.

$\hat{U} = \frac{1}{mag} \to$ direc

$\hat{U} = \frac{\vec{U}}{|\vec{U}|}$

$$\boxed{\vec{U} = U\hat{U}}$$

a b b

Shm, -

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$\vec{U} = 2\vec{a} - 3\vec{b} + \vec{c} = 2(\hat{i} - 2\hat{j} + \hat{k}) - 3(\ +\ ) + \ = $

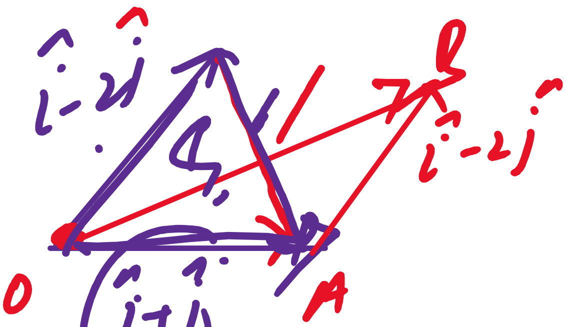
$|\vec{U}| = \sqrt{a^2 + b^2 + c^2}$

$\hat{U} = \frac{\vec{U}}{|\vec{U}|}$, $\vec{U} = 2\vec{a} - 3\vec{b} + \vec{c}$

$|U|$

Use vectors to find the length of diagonals of a parallelogram having adjacent sides $\hat{\imath} + \hat{\jmath}$ and $\hat{\imath} - 2\hat{\jmath}$.

$\vec{OB} = \vec{OA} + \vec{AB}$

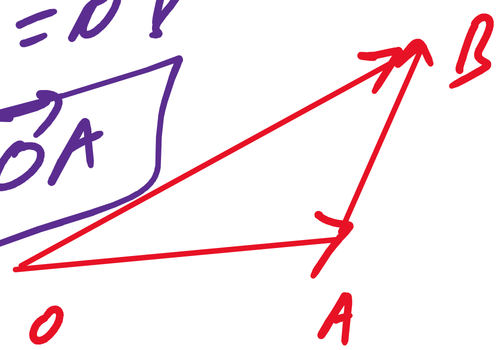


$\vec{OB} = 2\hat{i} - \hat{j}$

$|\vec{OB}| = \sqrt{4+1} = \sqrt{5}$

$\vec{OA} + \vec{AB} = \vec{OB}$

$\boxed{\vec{AB} = \vec{OB} - \vec{OA}}$



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$\vec V = \vec a - 2\vec b$, $\hat V = \frac{\vec V}{|\vec V|}$

$-2\hat i + 6\hat j + 10\hat k$

$\hat V = \vec a - 2\vec b$

8. If $\vec a = 2\hat\imath - 3\hat\jmath + \hat k$ and $\vec b = \hat\imath - 3\hat\jmath + 5\hat k$ then find:

(i) A vector of magnitude 5 in the direction $\vec a - 2\vec b$.

(ii) A vector of magnitude $\frac{3}{7}$ opposite in direction $3\vec a + \vec b$.

9. The position vectors of points A and B are $\hat\imath - 2\hat\jmath + \hat k$ and $2\hat\imath + 3\hat\jmath - \hat k$ respectively.

(i) Find the position vector of point P dividing the line segment joining A and B in the ratio 2 : 3 internally.

(ii) Find the position vector of point Q dividing the line segment $\overline{AB}$ in the ratio 3 : 2 externally.

$\vec V = 3\hat i + 11\hat k$

$\frac{2\cdot 3 + 3 \cdot 1}{2+3}, \frac{2\cdot 6 - 3 \cdot 9}{2-3}$

$\sqrt{9+121}$

$\hat V = \frac{\vec V}{|\vec V|} = \frac{3\hat i + 11\hat k}{\sqrt{130}}$

$\vec a - 2\vec b$

i) $\vec U = |U|\,\hat U$

$\boxed{\vec U = 5\hat V}$

$\vec V = 3\vec a + \vec b$

$\hat V = \frac{\vec V}{|\vec V|}$

$\vec U = U\hat U$

3

$$\hat{u} = -\frac{3}{7}\hat{v}$$

Example 9: Find the coordinates of point P, if $\overrightarrow{OP}$ is a vector of magnitude 2 and is parallel to the vector $2\hat{\imath} - 3\hat{\jmath} + 4\hat{k}$.

$\vec{v} = 2\hat{i} - 3\hat{j} + 4\hat{k}$

$\overrightarrow{OP} = 2 \cdot \hat{v}$

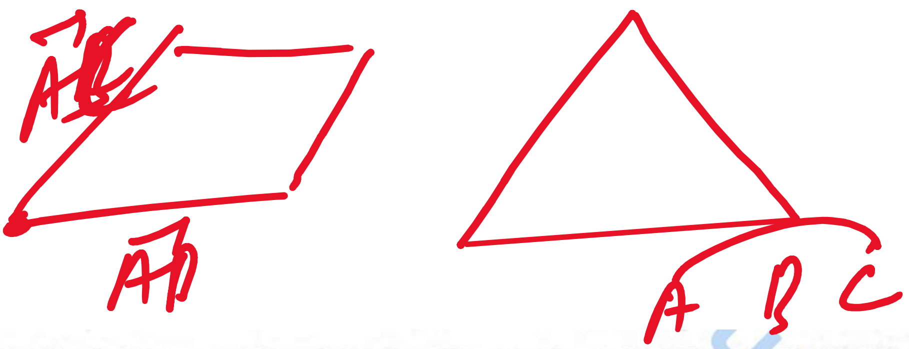
$\hat{v} = \dfrac{2(2\hat{i} - 3\hat{j} + 4\hat{k})}{\sqrt{4+9+16}}$

$\left(\dfrac{4}{\sqrt{29}}, \dfrac{-6}{\sqrt{29}}, \dfrac{8}{\sqrt{29}}\right) = 2 \, \dfrac{\vec{v}}{\sqrt{4+9+16}} = \dfrac{2}{\sqrt{29}}$

Example 10: Two direction angles of vector $\vec{r}$ are $30°$ and $60°$. Find the third direction angle. Also find unit vector $\hat{r}$.

Example 13: Prove that $\cos(\alpha - \beta) = \cos\alpha \cos\beta + \sin\alpha \sin\beta$

Example 14: For any triangle ABC, with usual notations prove that $|\vec{a}| = |\vec{b}|\cos\gamma + |\vec{c}|\cos\beta$



Example 17: Find the area of a triangle with vertices (0, 0), (2, 9), (3, 5).

$\vec{AD}$ 1 $|\vec{AB} \times \vec{AC}|$

$\bar{A}D = \frac{1}{2} |\vec{AB} \times AC|$

Example 16: If $\vec{a} = 2\hat{\imath} + 5\hat{\jmath} - 2\hat{k}$ and $\vec{b} = \hat{\imath} - \hat{\jmath} + 3\hat{k}$ are two adjacent sides of a parallelogram, then find its area.

Example 18: If $\vec{a} = 2\hat{\imath} - \hat{\jmath} + \hat{k}$; $\vec{b} = -\hat{\imath} + 2\hat{\jmath} - \hat{k}$ and $\vec{c} = \hat{\imath} + \hat{\jmath} - 3\hat{k}$ then find the angle between the vectors $\vec{a} + \vec{b}$ and $\vec{a} + \vec{c}$.

Example 19: Show that $\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$.

Example 20: Find the moment of the force $\vec{F} = 3\hat{\imath} - 2\hat{\jmath} + 5\hat{k}$ about the point $(2, 1, -1)$ when it is applied at point $(3, 0, 2)$.

Example 26: Find the volume of a tetrahedron with vertices $A(0, 0, 0)$, $B(1, 3, -1)$, $C(2, 2, 1)$ and $D(1, 6, 5)$.